



POST AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY



Fundamental Concepts

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- What is an embedded system?
- Characteristics of embedded systems.
- Requirements of embedded systems.
- General model of embedded systems.
- Classification of embedded systems



What is an embedded system?

- ❑ An embedded system is a term used to describe a system capable of autonomous operation that is integrated or ‘embedded’ within a larger and more complex environment or system.
 - According to the IEEE definition, it is a computing system that resides within another product and is typically hidden from the end user.
 - Key characteristics:
 - Integration of hardware and software
 - Application-specific nature
 - Resource constraints
 - Responsiveness and real-time behavior
 - High reliability

❖ What is a system?

- A system is an organized arrangement of functional units that work together to perform tasks.
- Its purpose is to organize or execute one or more tasks based on a set of rules, plans, or programs.
- Examples: clocks, washing machines, etc.

❖ Some definitions of embedded systems:

- A system in which hardware is integrated with software to perform a specific application, or a part of an application within a larger system.
- A system in which software with a specific function is embedded into computer hardware.
- A hardware system designed for a specific product or application. It may operate as a standalone system or as part of a larger system, with the software embedded in ROM or flash memory.

❖ Main components of an embedded system:

- Hardware components: ROM, RAM, EEPROM, I/O interfaces, processor, power supply, etc.
- Software components
- Real-time operating system (RTOS) (sometimes considered part of the software component)
- Environment interaction components: sensors and actuators

❖ Main components of an embedded system:

■ Hardware components:

● **Central Processing Unit (CPU):**

- The most important functional block, responsible for data processing and controlling other components.
- It can be a standalone microprocessor or a microcontroller that integrates the CPU, RAM, ROM, and I/O on a single chip to save space and reduce cost.

● **Memory:**

- ROM (Read-Only Memory): stores permanent program code
- RAM (Random Access Memory): stores temporary data and computation results.

● **Bus system:** A collection of signal lines (data, address, and control buses) that connect the CPU with memory and interface circuits.

❖ Main components of an embedded system:

- Hardware components:
 - Input/Output (I/O) interfaces: Allow the CPU to exchange data with peripheral devices such as keyboards, displays, and sensors.
 - ADC and DAC converters:
 - ADC (Analog-to-Digital Converter): Converts analog signals from the environment into digital signals for processing;
 - DAC (Digital-to-Analog Converter): Performs the reverse operation to control actuators.
 - System clock: Provides a reference clock to synchronize all operations of the integrated circuits on the board.
 - Power supply unit: Supplies power to the entire system, typically implemented using a switching power supply or batteries/accumulators for portable devices.

❖ Main components of an embedded system:

- Software components
 - Device drivers: The lowest software layer, which directly interacts with hardware to initialize and manage access to hardware resources.
 - Real-time operating system (RTOS): Manages system resources, schedules tasks, and ensures that the system responds correctly to events under strict real-time constraints.
 - Middleware: Provides shared services for applications, such as networking protocols (e.g., TCP/IP), data security, or a Java virtual machine.
 - Application layer: The top layer that defines the specific functions executed by the embedded system (e.g., control software for a washing machine or a mobile phone).

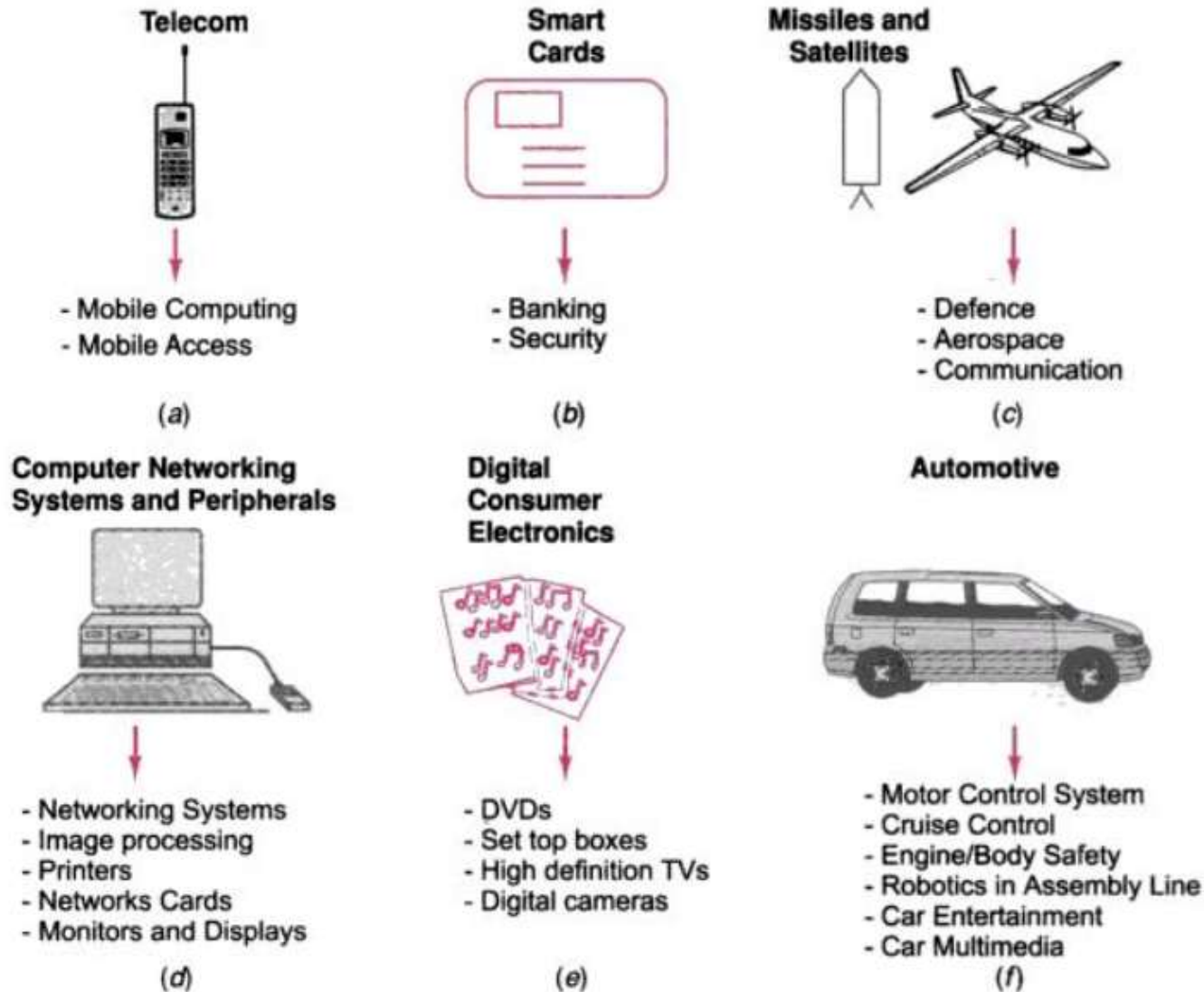
What is an embedded system?

❖ Main components of an embedded system:

- Environment interaction components
 - Sensors: Capture information from the environment such as temperature, speed, and position.
 - Actuators: Receive commands from the system to perform physical actions such as opening/closing valves or driving motors.

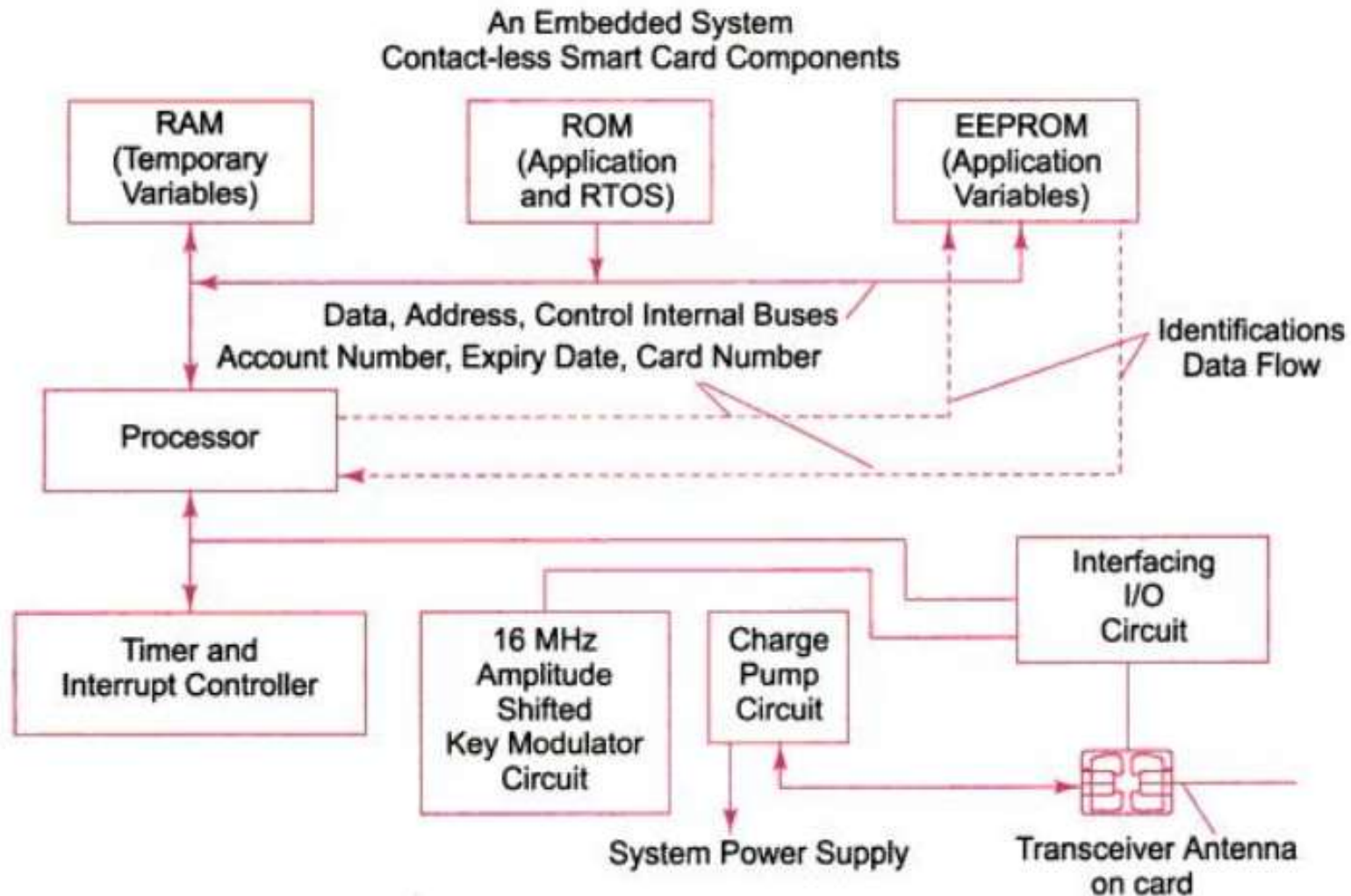
What is an embedded system?

❖ Some examples of embedded systems:



What is an embedded system?

❖ Some examples of embedded systems:



❖ **An embedded system typically requires:**

- Dedicated functional tasks
- Dedicated and complex algorithms
- User interfaces
- Real time operations
- Multi-rate operation: tasks with different processing rate requirements (e.g., audio, video, etc.)

❖ **An embedded system must also meet requirements related to:**

- Memory capacity
- Processing speed
- Low power consumption

❖ Important criteria to consider:

- Performance
- Power consumption
- Size
- Design cost
- Packaging cost

❖ Key characteristics of embedded systems:

■ Limited hardware and software resources:

- An embedded system is a special type of computer that is significantly constrained in terms of processing performance, memory capacity, and power consumption compared to general-purpose computers such as PCs or servers.
- The software is typically fixed or specially designed, with few high-level abstraction layers in order to optimize the limited storage capacity of ROM.

■ Application-specific nature:

- Unlike general-purpose computers, embedded systems are designed to perform only one or a limited number of specific applications.

❖ Key characteristics of embedded systems:

- **Reactivity and environmental interaction:** Embedded systems can continuously interact with the application environment through sensors. They are reactive intelligent devices capable of providing timely responses to events arising from the environment.
- **Autonomous and reliable operation:** Embedded systems typically operate autonomously with very high requirements for reliability, availability, and safety. In the event of a failure, the system must ensure that no harm is caused to the overall system.
- **Tolerance to harsh environments:** Embedded systems require very high quality to operate reliably in harsh conditions, such as extremely high or low temperatures, high humidity, strong vibrations, and severe electromagnetic interference.

❖ Key characteristics of embedded systems:

■ **Real-time constraints:**

- Most embedded systems must satisfy strict timing requirements, meaning that outputs must be correct and produced at the appropriate time relative to input events.
- This includes both time-sensitive and time-critical behavior.

■ **High performance and optimization:** Embedded systems require good processing performance while ensuring low power consumption, optimized software code size, small physical footprint, and competitive cost.

❖ Embedded systems can be classified based on various criteria

- Classification by operating environment
 - Standalone systems: These systems receive inputs from controlled entities, process them, and produce responses within a specified time frame according to the design objectives.
 - Networked systems (networked embedded systems): Systems that are interconnected with each other or with a central control unit via local area networks or wide area networks (e.g., mobile communication network systems).

❖ Embedded systems can be classified based on various criteria

- Classification by architecture and scale
 - Small-scale systems: Typically designed with a single 4-bit or 8-bit CPU, simple hardware, simple software, and very low power consumption.
 - Medium/complex-scale systems: Use 16-bit or 32-bit CPUs, feature extended bus architectures for peripheral interfacing, and employ sophisticated embedded software, often running a real-time operating system (RTOS).
 - Sophisticated/large-scale systems: Require highly specialized hardware and software, may use multiple CPUs or programmable logic devices (PLA/FPGA), and are commonly applied in aerospace, military, or high-end networking equipment.

❖ Embedded systems can be classified based on various criteria

■ Classification by hardware platform

- Microprocessor-based embedded systems: Use a standalone CPU combined with discrete RAM, ROM, and I/O chips on the board. This type offers high computational capability but is more expensive and has a larger physical size.
- Microcontroller-based embedded systems: All components (CPU, RAM, ROM, I/O, timers) are integrated into a single chip, resulting in shorter design time, compact size, and lower cost.

❖ Embedded systems can be classified based on various criteria

- Classification by real-time characteristics and behavior
 - Hard real-time systems: Require extremely strict timing guarantees; missing a deadline may lead to serious system failures (e.g., automotive airbag systems, safety-critical medical devices).
 - Soft real-time systems: Meeting deadlines is desirable to maintain performance, but occasional delays do not cause severe consequences.
 - By response mechanism: Systems can be classified into event-triggered and time-triggered systems.

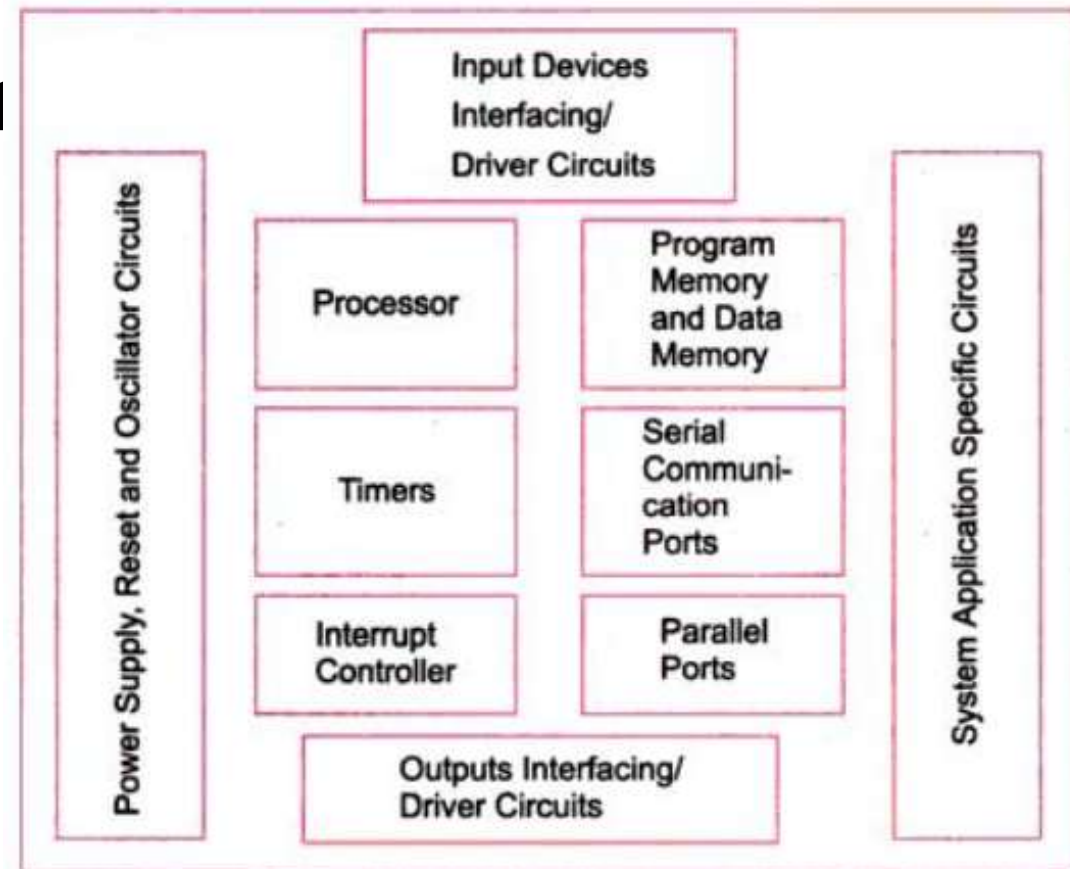
❖ Embedded systems can be classified based on various criteria

■ Classification by generation of development

- First generation: Based on simple 8-bit microprocessors (such as the 8085 and Z80), with software written in assembly language.
- Second generation: Utilized 16-bit microprocessors, with the emergence of embedded operating systems.
- Third generation: Transitioned to 32-bit microprocessors, with support for specialized processors such as DSPs or ASICs.
- Fourth generation: Employs System-on-Chip (SoC) technology, multi-core processors, and very high performance (e.g., smartphones).

❖ The general model of an Embedded System can essentially be explained from three main perspectives:

- Hardware functional blocks
- Mechanisms of interaction with
- Layered software architecture



❖ The general model of an Embedded System can essentially be explained from three main perspectives:

- Basic Hardware Functional Blocks
 - Central Processing Unit (CPU): The “brain” that decodes instructions and processes data according to the pre-installed program.
 - Memory: Includes ROM for storing permanent program code and RAM for storing data and intermediate results during computation.
 - Bus System: A collection of data, address, and control lines that connect the CPU with memory and peripheral circuits.
 - Input/Output (I/O) Interface: Enables the CPU to exchange data with external devices through communication standards.
 - Support Blocks: Include the System Clock for synchronizing operations and the Power Supply that provides energy for the entire system.

❖ The general model of an Embedded System can essentially be explained from three main perspectives:

- Environment Interaction Model

- Sensors: Capture physical information from the environment (temperature, pressure, speed, etc.).
- ADC/DAC Converters: ADC converts analog signals from sensors into digital signals for CPU processing; DAC performs the reverse, converting digital commands into analog signals to control devices.
- Actuators: Execute specific physical actions according to commands from the system.
- Network Interfaces: Connect with other embedded systems or control centers (such as SCADA) via communication protocols.

❖ The general model of an Embedded System can essentially be explained from three main perspectives:

- Abstract Architecture (Layered Model)
 - Hardware Layer: The lowest layer, consisting of physical components.
 - System Software Layer: Includes the operating system (OS/RTOS) or control programs (Monitor), responsible for resource management and task coordination.
 - Application Layer: The top layer contains application programs that define the specific functions of the product.

The layered architecture is characterized by each layer using only the functions (or services) of the layer below it, while only allowing the layer above to use its functions (services). This architecture offers advantages in security, sustainability, simple design, ease of upgrading (services), modular implementation of each layer, and the ability to perform “hot” upgrades even while the system is running.

- ❖ The types of processors (CPUs) used in embedded systems are very diverse and are classified based on architecture, level of integration, and specific application purposes.
 - Classification by Level of Integration and Configuration
 - Classification by Specialized Processing Purpose
 - Classification by Instruction Set Architecture and Memory

- ❖ The types of processors (CPUs) used in embedded systems
 - Classification by Level of Integration and Configuration
 - **Microprocessor:**
 - These are standalone CPUs with strong computing capabilities but require external chips for RAM, ROM, I/O, and timing.
 - This type is usually a general-purpose CPU simplified to suit embedded applications.
 - Typical examples include the Intel 8080/8085 or Intel x86 (8086/8088) series.

- ❖ The types of processors (CPUs) used in embedded systems
 - Classification by Level of Integration and Configuration
 - **Microcontroller (MCU):** Designed with the all-in-one criterion, integrating CPU, RAM, ROM, I/O ports, and timers within a single chip.
 - The advantages of this type are compact size, low cost, and reduced design time.
 - Common families include Intel 8051/8052, PIC (Programmable Intelligent Computer), and various Atmel series.

- ❖ The types of processors (CPUs) used in embedded systems
 - Classification by Level of Integration and Configuration
 - **Programmable System on Chip (PSoC):**
 - This type of chip integrates a microcontroller with configurable analog and digital functional blocks.
 - An example is the CY8C29466 series from Cypress.

❖ The types of processors (CPUs) used in embedded systems

■ Classification by Specialized Processing Purpose

- Digital Signal Processor (DSP): Specifically designed to handle signal processing tasks such as audio, image, and video at high speed. Common examples include the TMS320 series from Texas Instruments and BlackFin chips.
- Application-Specific Integrated Circuit (ASIC): Chips custom-designed for a specific application according to particular requirements, achieving performance optimization far beyond general-purpose CPUs.
- Field Programmable Gate Array (FPGA): Provides reprogrammable logic structures, often used in advanced embedded systems that require high flexibility and parallel processing capability.

❖ The types of processors (CPUs) used in embedded systems

- Classification by Instruction Set Architecture and Memory
 - Instruction Set Architecture: Includes CISC (Complex Instruction Set Computing) with a full instruction set and RISC (Reduced Instruction Set Computing) with a simplified instruction set. RISC is very common in modern embedded systems, such as the ARM architecture, to optimize power consumption and processing speed.
 - Memory Access Architecture:
 - Von Neumann Architecture: Uses a common BUS system for both instructions and data, which prevents simultaneous transmission of both.
 - Harvard Architecture: Separates data memory (RAM) and program memory (ROM/Flash) with independent BUS systems, allowing the CPU to fetch instructions and access data simultaneously, thereby increasing processing speed. This architecture is very common in embedded microcontroller design.

- ❖ **General-Purpose Microprocessor** (also called a general-purpose CPU): A general-purpose microprocessor is a central processing unit (CPU) designed to execute a wide range of computational tasks and applications, rather than being limited to a specialized function.
 - **Independent Hardware Architecture:** Standalone CPU. To form a functioning system, it must be combined with external chips such as RAM, ROM, the BUS system, and input/output (I/O) interface circuits.
 - **High Computational Performance:** These processors prioritize processing speed and strong computing capability, often achieving billions of instructions per second (MIPS).
 - **Flexible Programmability:** Programs can be freely installed, removed, or modified according to user needs.
 - **Support for General-Purpose Operating Systems (GPOS):** Such as Windows, Linux, or Unix...
 - **Common Examples:** Processors like the Intel x86 series (Pentium, Core i7) or AMD A-series

❖ **Single Purpose Processor** (also called a specialized or dedicated processor) is a type of microprocessor designed to perform one or a few specific technical tasks or functions, rather than being a general-purpose processor.

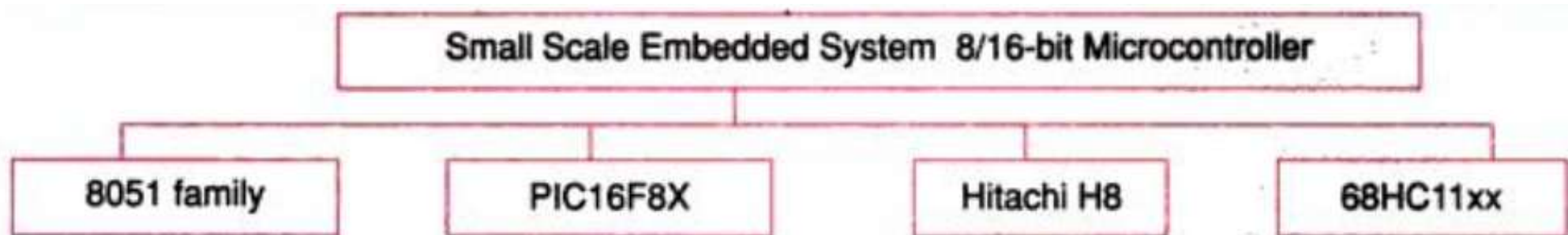
For example, ASIC:

- Floating point Coprocessor
- Pixel coprocessor
- Image codec in digital camera
- Graphic processor
- Speech processor
- Adaptive filtering processor
- Encryption engine
- Decryption engine
- Communication protocol stack processor

❖ ASIP (Application Specific Instruction-Set Processor):

■ Small-Scale Embedded Systems:

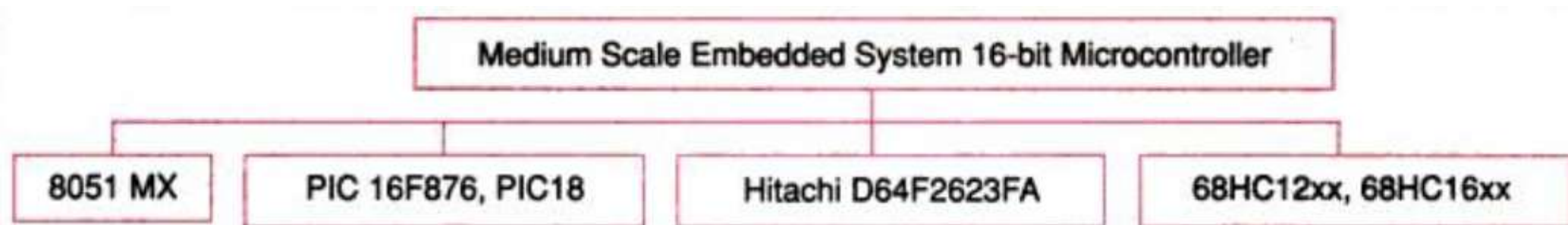
- Simple Hardware: A single CPU, typically 4-bit or 8-bit.
- Simple Software: Uses a control program (monitor) to manage operations; executable code is loaded into ROM with limited capacity.
- Low Power Consumption: Requires very little energy.
- Example Applications: Air conditioners, heaters, electronic toys.



❖ ASIP (Application Specific Instruction-Set Processor):

■ Medium-Scale Embedded Systems:

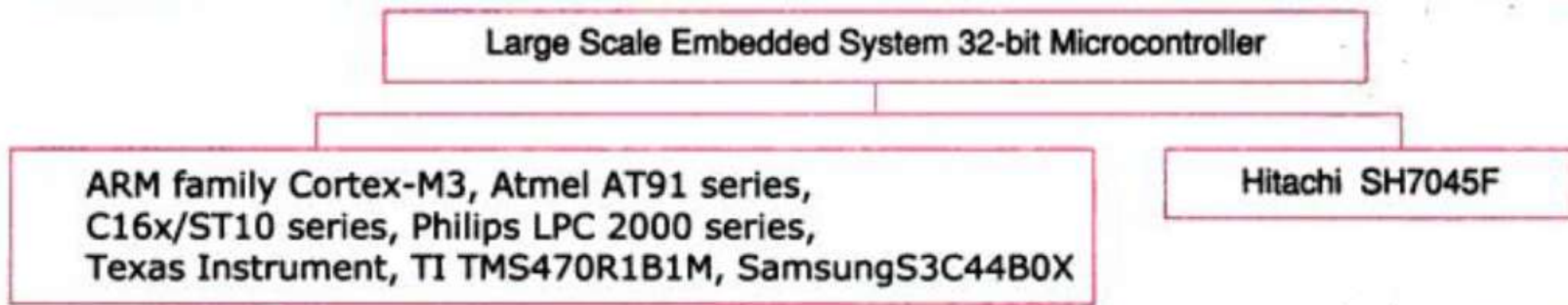
- These systems are usually designed based on 16-bit or 32-bit microprocessors or microcontrollers. They feature an extended BUS architecture to support connection with multiple peripheral devices.
- The system can be built using Digital Signal Processors (DSPs) to handle specialized tasks, such as metal cutting machines, plastic injection molding machines, data acquisition systems, and Supervisory Control and Data Acquisition (SCADA) systems.
- Embedded software is often sophisticated, requiring the ability to handle multiple tasks and concurrent operations. Medium-scale systems usually include an embedded operating system (GPOS or RTOS) for resource management and task scheduling.



❖ ASIP (Application Specific Instruction-Set Processor):

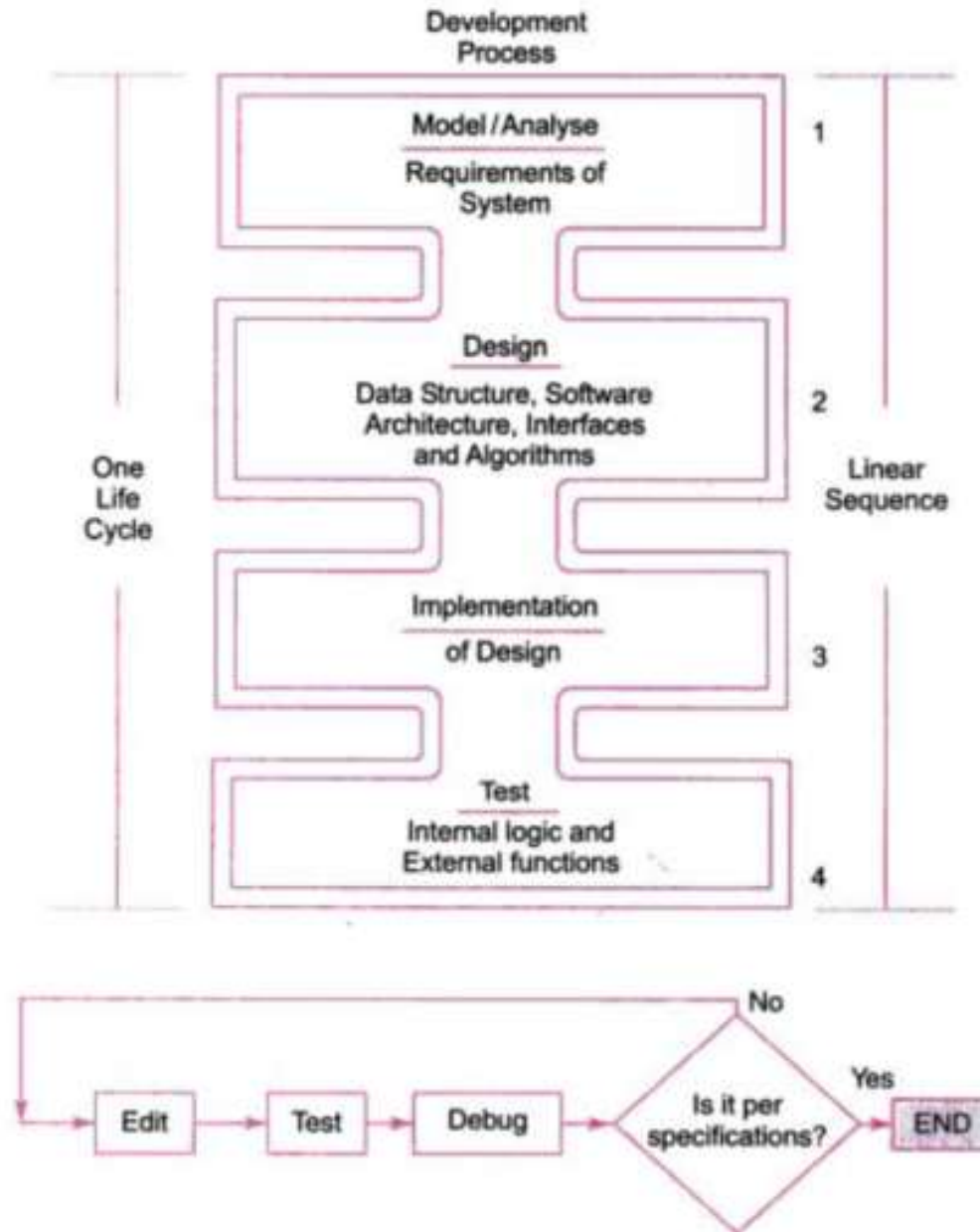
■ Large-Scale Embedded Systems:

- These systems often use multiple processors (multi-processor) operating in parallel or CPUs with reconfigurable capabilities
- Software functions go beyond simple control tasks and include complex encoding/decoding algorithms
- This type of system is governed by strict requirements for processing speed and absolute reliability, and is typically found in high-end networking devices, military and civilian avionics control systems, or advanced medical equipment.



- ❖ Multicore Processor is a core component in modern computer systems and embedded systems, representing a significant advancement in performance and integration.
 - Multicore processors are a hallmark of the fourth generation in the history of embedded system (ES) development. This technology provides high performance, tight integration, and the ability to miniaturize devices.
 - Application Scope: They are commonly used in large-scale embedded systems or sophisticated systems that require powerful computational capabilities.
 - Systems using multicore processors often require high-performance real-time operating systems (RTOS) to manage task scheduling, prioritize tasks, and coordinate resources precisely.

Design process



❖ Main Steps:

1. Identify the system objects/requirements
2. Define hardware and software architecture
3. Determine additional functions
4. Study related families and designs
5. Break the design into smaller modules (hardware, software)
6. Mapping
7. Design system interfaces
8. Fine-tune the system

❖ Các tiêu chí:

<i>Design Metrics</i>	<i>Description</i>
Power Dissipation	For many systems, particularly battery operated systems, such as mobile phone or digital camera the power consumed by the system is an important feature. The battery needs to be recharged less frequently if power dissipation is small.
Performance	Instructions execution time in the system measures the performance. Smaller execution time means higher performance. For example, a mobile phone, voice signals processed between antenna and speaker in 0.1s shows phone performance. Consider another. For example, a digital camera, shooting a 4M pixel still image in 0.5s shows the camera performance.
Process deadlines	There are number of processes in the system, for example, keypad input processing, graphic display refresh, audio signals processing and video signals processing. These have deadlines within which each of them may be required to finish computations and give results.
User interfaces	These include keypad GUIs and VUIs.
Size	Size of the system is measured in terms of (i) physical space required, (ii) RAM in kB and internal flash memory requirements in MB or GB for running the software and for data storage and (iii) number of million logic gates in the hardware.
Engineering cost	Initial cost of developing, debugging and testing the hardware and software is called engineering cost and is a one-time non-recurring cost.
Manufacturing cost	Cost of manufacturing each unit.
Flexibility	Flexibility in design enables, without any significant engineering cost, development of different versions of a product and advanced versions later on. For example, software enhancement by adding extra functions necessitated by changing environment and software re-engineering.
Prototype development time	Time taken in days or months for developing the prototype and in-house testing for system functionalities. It includes engineering time and making the prototype time.
Time-to-market	Time taken in days or months after prototype development to put a product for users and consumers.
System and user safety	System safety in terms of accidental fall from hand or table, theft (e.g., a phone locking ability and tracing ability) and in terms of user safety when using a product (for example, automobile brake or engine).
Maintenance	Maintenance means changeability and additions to the system; for example, adding or updating software, data and hardware. Example of software maintenance is additional service or functionality software. Example of data maintenance is additional ring-tones, wallpapers, video-clips in mobile phone or extending card expiry date in case of smart card. Example of hardware maintenance is additional memory or changing the memory stick in mobile computer and digital camera.